

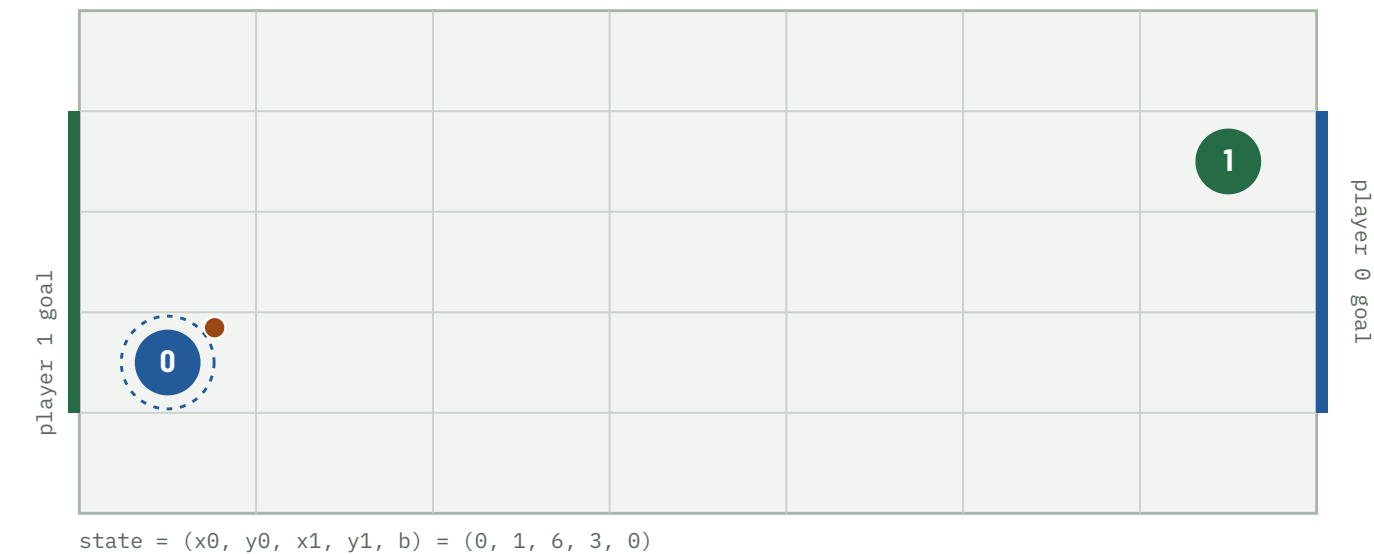
When Does Soccer Need Mixed Strategies?

A pure-first Nash-Q approach: where
a two-player soccer Markov game
does — and does not — need mixing,
and how much LP work that saves.

FOUR QUESTIONS

RQ1 Existence — under what transition structures does a pure equilibrium exist at every state? **RQ2 Efficiency** — how much work does checking for a pure saddle first eliminate? **RQ3 Mechanism** — which spatial configurations force mixing? **RQ4 Robustness** — how do discounting and tolerance affect the split?

repo · soccer-markov-nash | 210+ tests | game family in docs/design.md | methods in docs/methods.md | every policy drawn in docs/gallery.html



The kickoff. 7×5 grid, 2380 non-terminal states. Player 0 (blue) carries the ball and attacks the right edge; player 1 (green) the left. Start positions are the ones the A10 page assigns this netID. Actions U D L R are chosen simultaneously; reward is +1 / -1 / 0, zero-sum.

ABSTRACT

- 1** With deterministic move resolution — solved exactly, undiscounted, by 100-step backward induction — every one of the 238 000 (state $\times$ step) stage games has a pure saddle. A pure equilibrium exists; the LP is never called; discounting was never load-bearing. It is also positionally determined, with an explicit pure memoryless profile.

- 2** A coin-flip tie-break does not change this; Littman's random *move order* does — but only when it is the majority rule (a sharp threshold at $\text{blend} = 0.5$), only when the goal mouth exceeds one cell, and even then on 3.95% of states (7 $\times$ 5, 3-cell goal).

- 3** Those mixed states are geometrically localised (a decision tree predicts them at precision 0.96) and structurally uniform: 68 of 94 have a 2 $\times$ 2 matching-pennies support. The region is invariant to the kickoff and to the reward objective — it depends only on the resolution rule and the goal width.

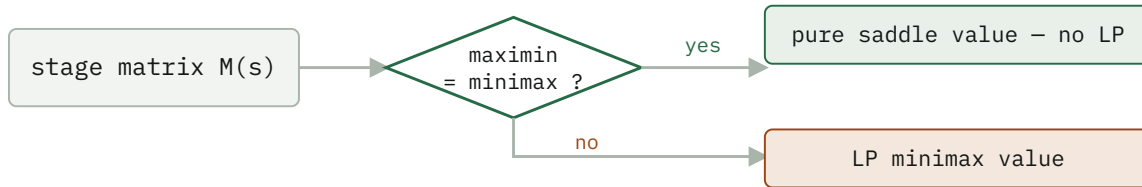
- 4** The pure-first hybrid solver matches the all-LP value function to 4e-16 while calling the LP on only 3.95% of states — $\sim 25\times$ fewer per sweep — though those states carry 41% of the equilibrium-path occupancy; mirror symmetry halves the work again.

Method

The game is zero-sum, so at every state the stage game is a payoff matrix $M(s)[a_0, a_1] = E[R + \gamma V(s')]$ and the Shapley (1953) fixed point is the minimax value:

$$\begin{aligned} V(s) = \text{val}(M(s)) &= \max_n \min_{k \in \mathcal{K}} (p^T M)[a_1] \\ &= \min_p \max_{a_0} (Mq)[a_0] \end{aligned}$$

The $Q = R + \beta \text{Nash}(Q')$ update is this operator. The A10 game is *undiscounted* (± 1 at a goal, tie after 100 steps); $\gamma < 1$ is a contraction device for value iteration and RQ4 shows the split holds across γ . Three stage backups: **pure** (the maximin security value — a fast lower bound, not a Nash value when it is below the minimax), **mixed** (the LP minimax value, via scipy HiGHS), and **hybrid** (the pure saddle value where maximin = minimax, the LP elsewhere — exact everywhere). Support enumeration (support_enum.py) finds *all* equilibria, including of general-sum games.



The pure-first hybrid. On the deterministic game the "yes" branch is taken at every one of the 2380 states, so the LP is never called.

RQ1 – Existence

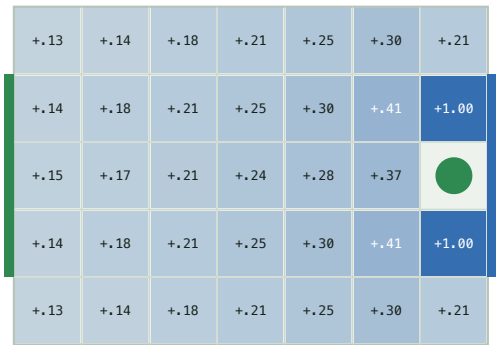
The A10 game is undiscounted, so it is solved exactly by 100-step backward induction (`run_finite_horizon`, `experiments/undiscounted.csv`):

EXACT UNDISCOUNTED GAME – 238 000 (STATE × STEP) STAGE GAMES

MOVE ORDER	MIXED STAGE GAMES	V(KICKOFF)	PURE EQUILIBRIUM?
deterministic (exact A10)	0	0.000	yes – pure (non-stationary)
coinflip tie-break	0	0.000	yes
random move order	3 148 (404 states)	+0.459	no

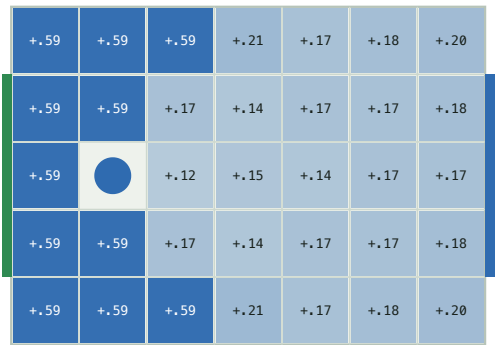
The deterministic equilibrium is **constructive and memoryless**: each player's win attractor (states from which it forces a goal, `soccer_nash/attractor.py`) equals its $V^* = \pm 1$ set, and the pure profile "attractor move on your winning set, safety move elsewhere" realizes V^* at every state (`scripts/positional.py`) — the game is positionally determined.

carrier position (player 0)



V^* by player 0 position (blue = player 0 ahead)

defender position (player 1)



V^* by player 1 position (blue = player 0 ahead)

The **value surface** of the random-order game ($\gamma = 0.9$). Left: V^* by carrier position — a smooth gradient toward the attacking goal, saturating to +1 on the two cells that score in one move. Right: V^* by defender position with the carrier held near midfield — nearly flat, the defender's location barely moves the value.

The stationary $\gamma < 1$ solve agrees and is faster — 0 no-pure-saddle stage games on the deterministic game at every γ from 0.5 to 0.995, and 94 / 2380 on the random game at $\gamma = 0.9$ ($V(\text{kickoff}) = +0.164$ at the netID start). The no-pure-saddle count never depends on the kickoff.

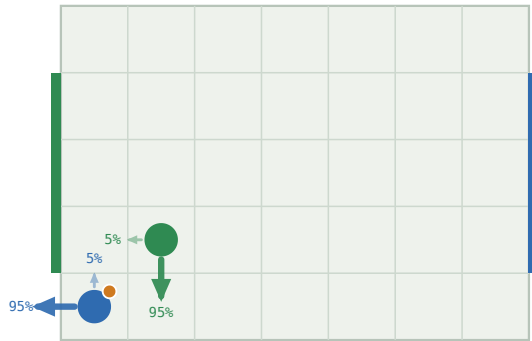
IT IS THE MOVE ORDER, NOT THE RANDOMNESS – AND IT HAS TO BE THE MAJORITY

The deterministic and coin-flip value functions are *identical* ($\max |V_{\text{det}} - V_{\text{coin}}| = 0$ at every γ): optimal play never enters a losing contest, so the coin is never flipped. Only Littman's random move order — where a ball-steal depends on who resolves first *and* on both players' targets — creates matching-pennies subgames. Interpolating the two (move_order="blend", resolve by random order with probability blend) shows the onset is **sharp: 0** mixed stage games for $\text{blend} \leq 0.5$, dozens at $\text{blend} = 0.52$, and an *overshoot* to $\sim 1.5\times$ the fully-random count near $\text{blend} \approx 0.7$ before it relaxes (scripts/blend.py, docs/blend.md). A minority of random-order resolution is washed out by the deterministic tie-break.



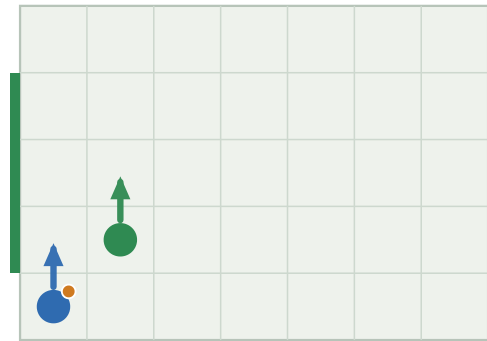
A threshold, not a ramp. No-pure-saddle stage games vs. the fraction of moves resolved by random order (5x4 board). Nothing until the random rule is the majority; then a jump and an overshoot.

random order -> mixed



mixed equilibrium · $V = +0.078$

deterministic -> pure



pure equilibrium · $V = +0.000$

One state, two resolution rules. Arrows are the equilibrium action distribution, sized by probability. Random order (Littman) makes the stage game matching pennies; deterministic carrier-wins collapses it to a pure best reply. Full set: [docs/gallery.html](https://docs.google.com/gallery).

RQ2 – Efficiency

Two numbers, kept apart: the **LP-call rate** is a property of the game and reproduces exactly; **wall clock** is a median of 5 repeats and depends on the machine and LP backend (`scripts/benchmark.py`).

RANDOM MOVE ORDER, $\Gamma = 0.9$ – THE CASE WHERE THE LP IS ACTUALLY NEEDED

SOLVER	SWEEPS	LP CALLS	STATES NEEDING LP	MEDIAN WALL CLOCK
pure (lower bound)	18	0	–	0.3 s
hybrid exact	108	9 672	3.95%	13.2 s
all-LP exact	108	~3.6 M	100%	~8 min

The hybrid solves an LP on only the 3.95% of states lacking a pure saddle — a $\sim 25\times$ per-sweep reduction — and matches the all-LP value to $4e-16$. The wall-clock ratio (~ 13 s vs ~ 8 min) is larger but less portable: it also reflects matrix construction and the LP backend. **Mirror symmetry** (`run_symmetric`, deterministic dynamics only): every state has a distinct board-flip-plus-player-swap mirror, so the 2380 states are 1190 pairs — reconstruct one from the other by $V(\text{mirror}(s)) = -V(s)$, cutting the deterministic solve 0.33 s $\rightarrow$ 0.23 s.

3.95% OF STATES, BUT 41% OF THE EQUILIBRIUM PATH

The LP-call rate counts the whole state space. Under the Nash policy from the kickoff only **456 of 2380** states are ever reached, and the 94 no-pure-saddle states carry **41%** of the discounted occupancy — a $2\times$ over-representation on the path, $10\times$ vs. the raw count (`soccer_nash/occupancy.py`, `docs/occupancy.md`). The carrier's optimal march to goal runs straight through the contested band; four of the eight most-visited states are mixed. The mixed structure is rare to *store* and common to *use*.

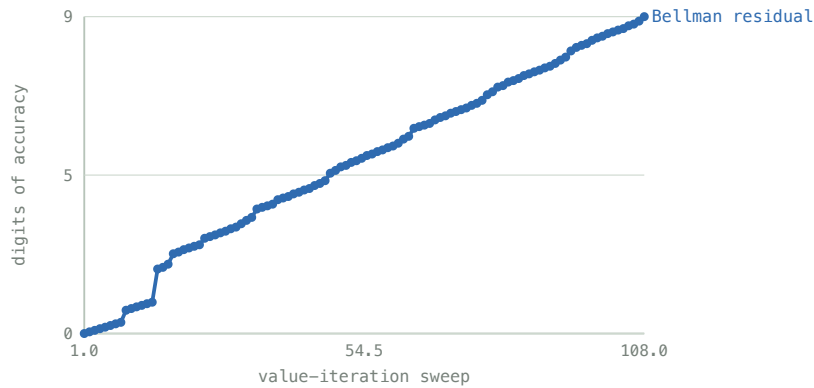
Freeze-then-iterate, and its staleness

RANDOM MOVE ORDER, HYBRID SOLVER – SAME FIXED POINT, $|\text{DIFF}| < 3E-9$

	ROUNDS / SWEEPS	MATRIX-GAME SOLVES	WALL CLOCK
value iteration	108	9 672	13.8 s
policy iteration	21	1 879	17.2 s

$5\times$ fewer LP solves — but the frozen strategies' distance to the current Nash stays *maximal* (a full pure-strategy flip) for **16 outer rounds**, then collapses to $< 1e-8$. The thrashing eats the saving; on the deterministic game it is strictly worse. Value iteration with the per-sweep Nash cache wins here.

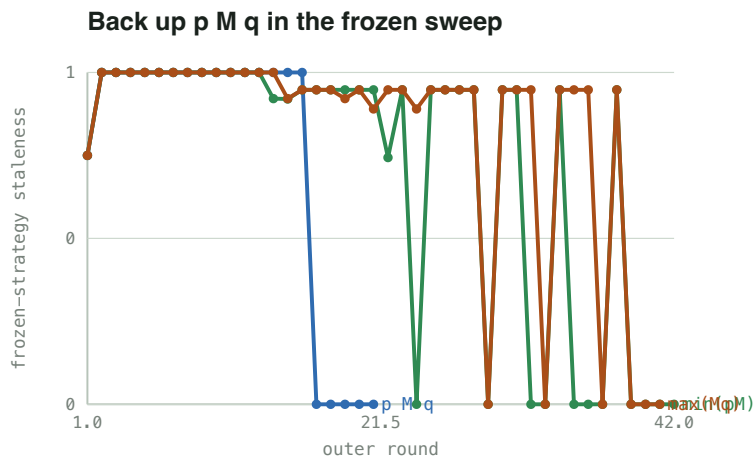
Value iteration: geometric decay at rate about gamma



Value iteration converges cleanly. The Bellman residual decays geometrically at rate $\approx \gamma - 108$ sweeps to $1e-9$. Freeze-then-iterate, by contrast, makes *no* progress until the frozen stage saddle flips (next figure) – concurrent stochastic games have no monotone policy improvement (docs/discussion.md §4).

WHICH VALUE QUANTITY TO BACK UP IN THE FROZEN SWEEP

The professor's specific question: the frozen (p, q) don't match the current Q , so does the evaluation sweep back up $p^T M q$, $\min_k (p M)_k$, or $\max_i (M q)_i$? All three reach the *same* fixed point (zero-sum equilibria are interchangeable), but **$p^T M q$ converges in half the rounds** (21 vs 41–42) — it is the unbiased estimate while the strategies are stale, where the two bounds are systematically pessimistic / optimistic and stretch the thrash phase from 16 rounds to 28 (`run_policy_iteration(eval_value=...)`, docs/numerics.md §3).

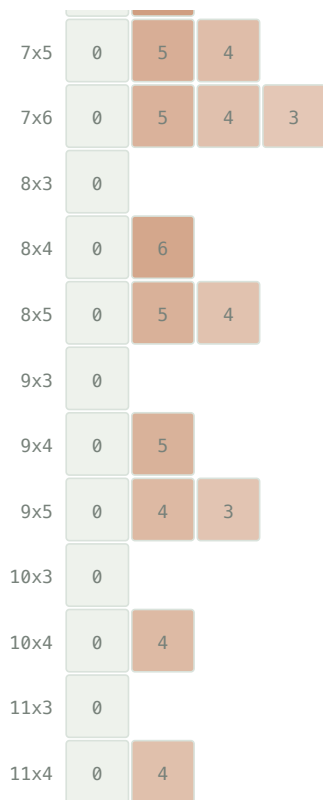


Back up $p^T M q$. Staleness of the frozen strategies per outer round. The middle quantity settles by round 16; the pessimistic $\min_k (pM)_k$ and optimistic $\max_i (Mq)_i$ keep flipping until round 28.

RQ3 – Mechanism

Goal-mouth width, not board size, is the gate. The phase diagram sweeps every board with width ≤ 11 , height ≤ 9 , width $\cdot$ height ≤ 45 against goal widths 1 to height-2 (scripts/phase_diagram.py --analyze, experiments/phase_diagram.csv).

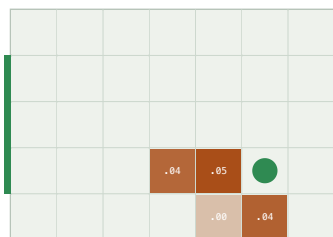
	goal-mouth width						
	1	2	3	4	5	6	7
3x3	0						
3x4	0	12					
3x5	0	10	9				
3x6	0	7	7	7			
3x7	0	7	6	6	6		
3x8	0	6	6	5	5	5	
3x9	0	6	5	5	4	4	4
4x3	0						
4x4	0	8					
4x5	0	8	6				
4x6	0	7	6	5			
4x7	0	7	5	5	4		
4x8	0	6	5	5	4	4	
4x9	0	6	5	4	4	4	3
5x3	0						
5x4	0	7					
5x5	0	6	4				
5x6	0	6	4	4			
5x7	0	5	4	4	3		
5x8	0	5	4	3	3	3	
5x9	0	5	4	3	3	3	3
6x3	0						
6x4	0	8					
6x5	0	6	4				
6x6	0	5	4	3			
6x7	0	4	4	4	3		
7x3	0						
7x4	0	7					



cells = mixed-state %

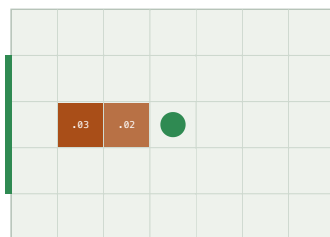
The gate. Goal width 1 → 0 mixed states, on all 40 one-cell configs. Goal width ≥ 2 → mixing on all 87: whether a board mixes is a *perfect* function of goal width. The *fraction* (2.7–12%) is an OLS R^2 0.64 in board area – a dilution effect – with goal width adding little and a negative coefficient. Every state is reachable from the kickoff.

defender at (5, 1)



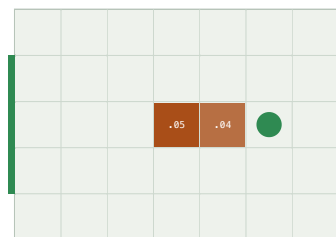
cells: minimax minus maximin of the carrier's stage game

defender at (3, 2)



cells: minimax minus maximin of the carrier's stage game

defender at (5, 2)



cells: minimax minus maximin of the carrier's stage game

Where the carrier has to guess. Defender pinned at the green disc; each cell shaded by minimax – maximin of the carrier's stage game (0 = pure saddle). Mixing is a local phenomenon – a handful of cells where the defender contests the forward lane.

Geometry predicts the label (scripts/geometry_model.py): a depth-4 decision tree on carrier-frame features separates mixed from the rest with **precision 0.96, recall 0.91**. The dominant feature is defender_can_intercept — the defender within one move of the carrier's forward cell ($P(\text{mixed}) = 0.44$ vs 0.002).

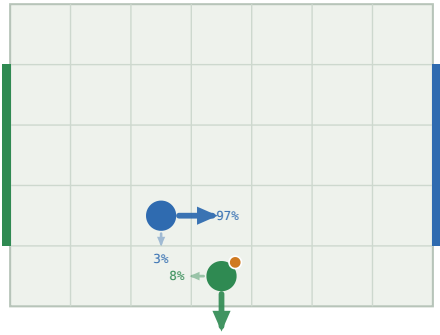
Beyond prediction, the 94 mixed states canonicalize under the board mirror to 47 pairs and cluster into 8 **geometric templates** (scripts/templates.py, docs/templates.md).

THE MIXED-STATE TEMPLATES – CARRIER-FRAME GEOMETRY, EQUILIBRIUM SUPPORT

STATES	GEOMETRY	SUPPORT	STRUCTURE
24	defender 1 ahead, carrier one row off the goal row	2×2	matching pennies
24	defender 1 ahead, same row, carrier on a goal row	2×2	matching pennies
16	defender 2 ahead, same row, carrier on a goal row	2×2	matching pennies
4	defender diagonally adjacent, carrier off the goal row	2×2	matching pennies
14	defender 1–2 ahead, same row, on a goal row	2×1	near-pure saddle
8	defender 2 ahead, same row, on a goal row	3×2	degenerate (row ties)
4	defender 2 ahead, carrier off the goal row	3×3	wider mix

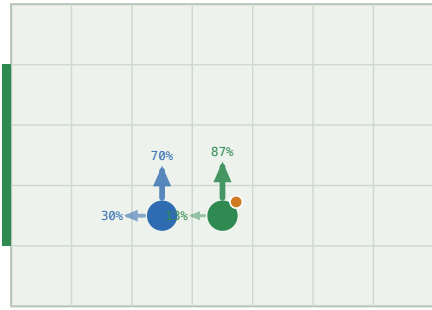
The four 2×2-support templates are all verified matching pennies — the carrier's best reply flips between the two defender columns and vice versa — covering 68 of 94 states: carrier {climb, advance} against defender {cover a lane, hold the forward cell}. All 94 share one value across every equilibrium; 64 have a unique one.

24 states $p^*=0.92$ $q^*=0.03$



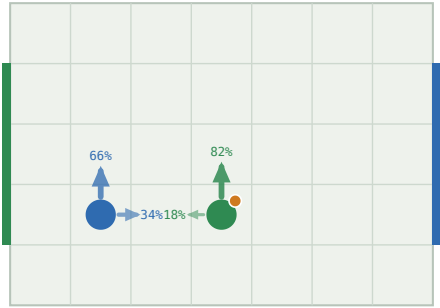
mixed equilibrium · $V = -0.150$

24 states $p^*=0.87$ $q^*=0.70$



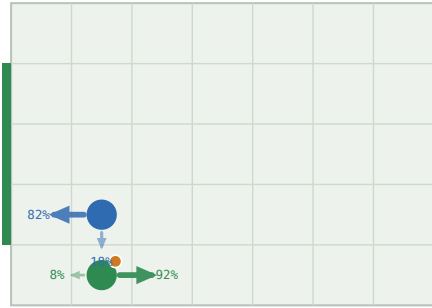
mixed equilibrium · $V = -0.199$

16 states $p^*=0.82$ $q^*=0.66$



mixed equilibrium · $V = -0.225$

4 states $p^*=0.08$ $q^*=0.18$



mixed equilibrium · $V = -0.206$

One policy fan per matching-pennies template. Each covers a class of mixed states related by carrier-frame geometry; p^* , q^* are the equilibrium weights on the first support action. The carrier randomizes between two scoring lanes, the defender between covering them.

Why the mix is unavoidable. A mixed stage game needs all three of: a stochastic ($\frac{1}{2}$ - $\frac{1}{2}$) resolution order; a goal mouth wider than one cell, so the carrier has two scoring lanes; and a defender close enough to contest the forward cell but unable to cover both lanes in one move. The carrier climbs or drops, the defender guesses which — matching pennies. Drop any one clause — a single goal cell, deterministic resolution, or the coin-flip tie-break — and every stage game has a pure saddle. The full argument and one stage matrix per template are in docs/templates.md. For the single goal cell the theorem is partly proved (docs/proof.md): the defender has a closed-form optimal strategy — guard the one cell — and every stage game is dominance-solvable, machine-checked for all boards up to 11×5 .

SHALLOW MIXES, BUT THEY COMPOUND

Every one of the 94 no-pure-saddle stage games sits within 0.07 of a pure saddle (minimax – maximin, median 0.029) — the equilibria are shallow, and the exact mixing weights are numerically delicate. What is robust: 68 have a 2×2 support and all 68 are structurally matching pennies, and the *value of mixing* — $V(\text{hybrid}) - V(\text{pure maximin})$ — is +0.13 on average at those states. It compounds along a path: at the centred kickoff a pure-strategy player secures only a draw (0.000) where mixing is worth +0.15 (scripts/mixing.py, docs/mixing.md).

The reward objective does not move the mixed region

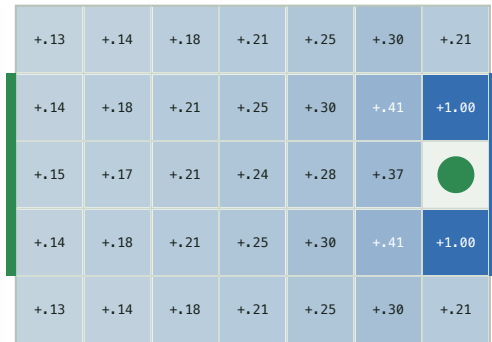
The default game rewards a *win* — first goal ends it, value $P(\text{win}) - P(\text{loss})$. A second objective, `scoring="rate"` (scripts/reward.py), keeps playing: a goal scores ± 1 and restarts with the ball to the conceding team, so the value is the expected discounted *goal difference* and $\gamma < 1$ is load-bearing. On the 7×5 random-order board:

SAME DYNAMICS, TWO REWARD OBJECTIVES – RANDOM MOVE ORDER, DISCOUNT 0.9

OBJECTIVE	NO-SADDLE STATES	VALUE RANGE	V(KICKOFF)
win – $P(\text{win}) - P(\text{loss})$	94	[−1.00, +1.00]	+0.150
rate – expected goal difference	94 – the same 94	[−0.88, +0.88]	+0.132

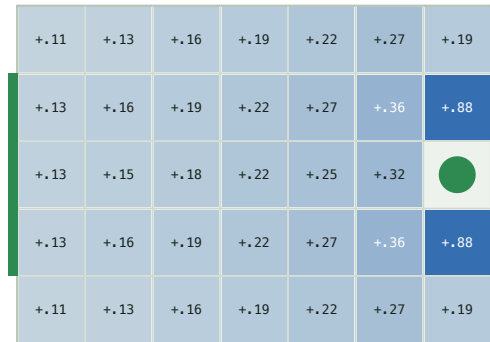
The no-pure-saddle set is *identical* — 0 states enter or leave. A mixed Nash equilibrium is an equilibrium of a single stage game $M(s)$; changing how the horizon is scored rescales $M(s)$ through $V(s')$ but does not cross the $\text{maximin} = \text{minimax}$ boundary. The value *range* compresses ($\pm 1 \rightarrow \pm 0.88$): under *rate* a lost position is not a cliff — conceding restarts play with possession, a floor under every state. Details in docs/reward.md.

scoring = win ($P(\text{win}) - P(\text{loss})$)



V* by player 0 position (blue = player 0 ahead)

scoring = rate (expected goal difference)



V* by player 0 position (blue = player 0 ahead)

Same shape, compressed range. V* by carrier position under the two objectives. rate pulls the goal-adjacent +1 cells down to +0.88 – the concede-and-restart floor – without changing where the value gradient runs.

Littman's fifth action

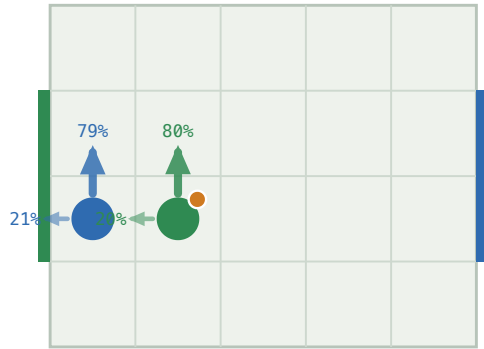
Littman's 1994 soccer game has a fifth action, stand, and his Figure 2 — a carrier pinned against its own goal by an adjacent defender — is the textbook "soccer needs mixing" example: the carrier must randomize between standing and moving. Adding stand to the random-order game (`n_actions=5`, `scripts/littman.py`) on Littman's 5×4 board:

LITTMAN'S BOARD (5×4, 2-CELL GOALS), RANDOM MOVE ORDER, DISCOUNT 0.9

ACTION SET	NO-SADDLE STATES	STAND IN THE MIXED SUPPORT	V(KICKOFF)
N, S, E, W	56	0	+0.147
N, S, E, W, stand	56	48	+0.172

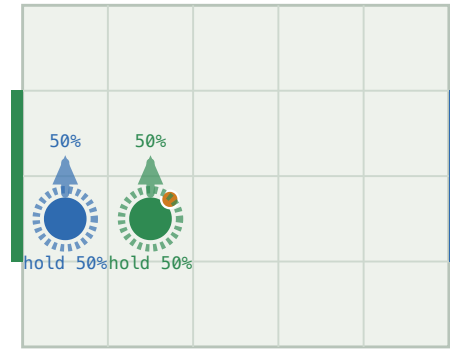
The fifth action does *not* move where mixing happens — 48 of the 56 no-pure-saddle states are shared, 8 swap in and 8 out at the margin — but it changes *what* the mix is: the carrier now hedges with stand in 48 of 56 states rather than with retreat, and every player is slightly better for the option ($V(\text{kickoff}) +0.147 \rightarrow +0.172$). The Figure 2 state is a clean matching-pennies matrix with equilibrium ($\frac{1}{2}$ climb, $\frac{1}{2}$ hold). The deterministic game keeps a pure saddle at every state with five actions too.

four moves



mixed equilibrium · $V = -0.255$

+ stand (Littman Fig 2)



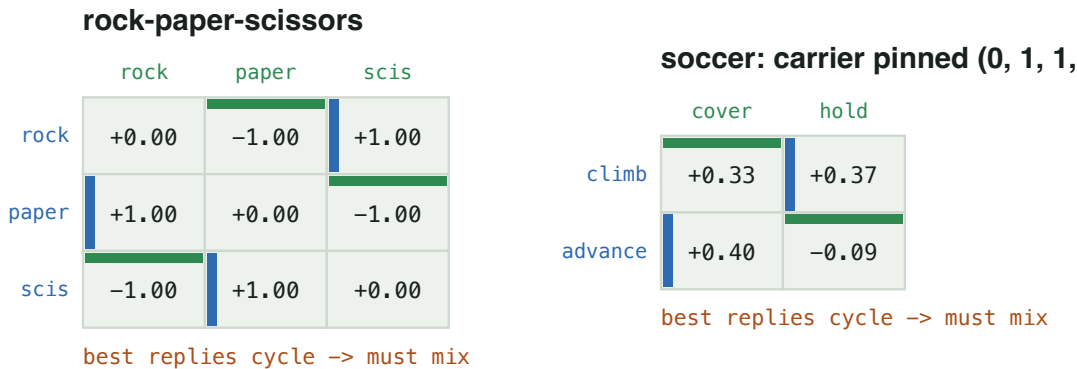
mixed equilibrium · $V = -0.315$

Littman's Figure 2. Same state, four moves (left) vs. five (right). The dashed ring is stand. Adding it turns the carrier's climb/retreat hedge into climb/hold – Littman's mixed policy.

RQ4 – Numerical robustness

The soccer game contains rock-paper-scissors

The pure/mixed check is the professor's $O(A^2)$ best-reply method: mark player 0's best row in each column and player 1's best column in each row; a cell with both is a pure saddle, and if none has both the best replies *cycle* and the game needs mixing — no LP required.



Same structure. Left: rock-paper-scissors. Right: the stage game at (0, 1, 1, 1, 1) — carrier pinned against its own goal, defender adjacent (Littman's Figure 2 geometry). Blue bar = player 0's best row per column; green bar = player 1's best column per row. They cross both ways — matching pennies, no pure equilibrium.

The value is three numbers

value_bracket returns what the row player guarantees ($\min_k (pM)_k$), gets ($p^T Mq$), and can reach ($\max_i (Mq)_i$) — the three quantities the professor flagged as never numerically identical. The true value is bracketed, so the width certifies the LP error.

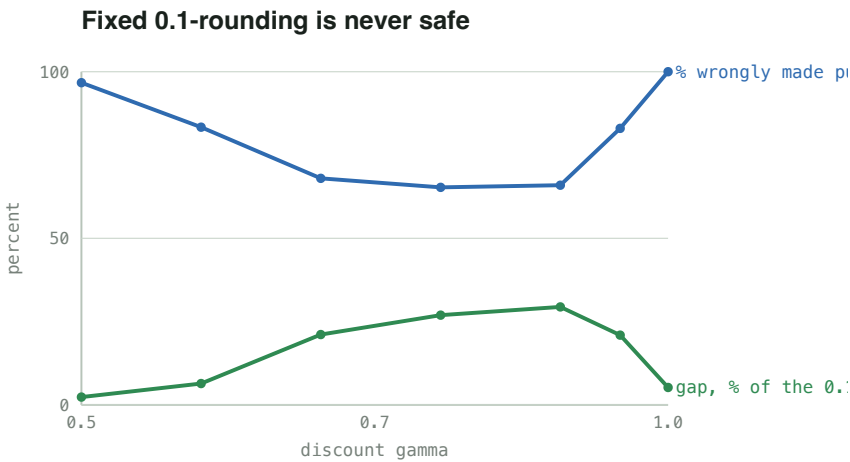
FINDING

With scipy's HiGHS the three agree to $1.1e-16$ per stage game, $8.7e-10$ accumulated — the definitional ambiguity is not a practical problem. If it were (an older vertex solver, or accumulated drift), the row player should take the *guaranteed* value $\min_k (pM)_k$: it is always a safe lower bound on the true minimax.

Discounting and tolerance

	DETERMINISTIC	RANDOM (7×5)
mixed states, $\gamma = 0.5 \rightarrow 0.9 \rightarrow 0.995$	0 → 0 → 0	122 → 94 → 102
classification split, $\text{rel_tol } 1\text{e-}12 \rightarrow 1\text{e-}3$	stable	604 / 1682 / 94 (stable)
classification, $\text{rel_tol } 1\text{e-}2 / 1\text{e-}1$	—	80 / 0 mixed
fixed 0.1-rounding flips a saddle	0	62

Discounting mildly shifts which states mix; the classification is otherwise robust across nine orders of magnitude of tolerance, breaking only as the tolerance approaches the real minimax – maximin gaps. Fixed 0.1-rounding — proposed to force indifference — misclassifies 62 states, exactly the small-entry mixed region.



There is no safe discount. The genuine minimax – maximin gaps (green) never reach the 0.1 rounding grid at any γ : heavy discounting shrinks the γ^* value differences, light discounting leaves entries barely apart. Fixed 0.1-rounding wrongly collapses 60–100% of the mixed games (blue). The scale-aware `rel_tol` classifier is stable across the whole range.

Secondary validation – the policy plays correctly

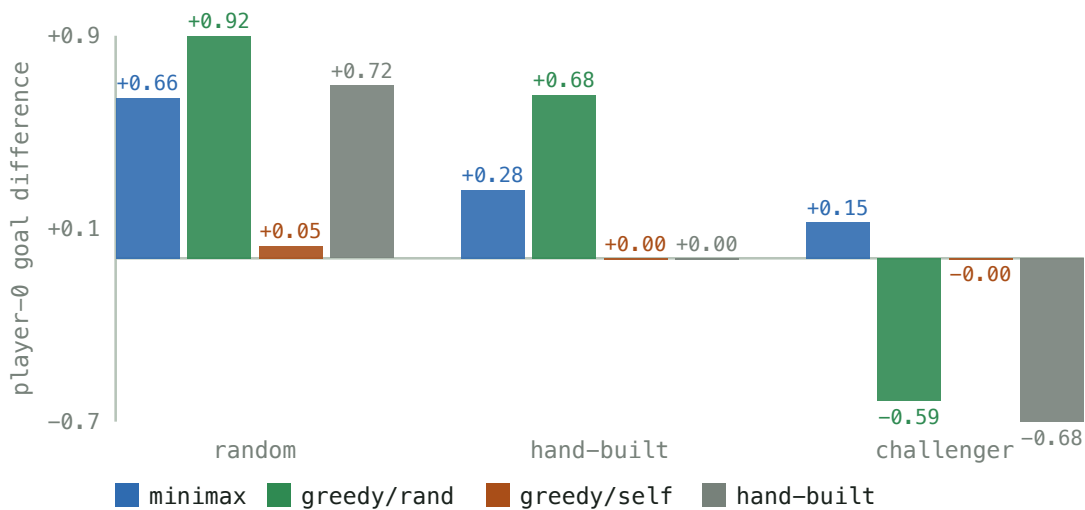
Littman's Table 3: minimax exploits *and* survives

Littman's central experiment trains four policies and scores each against a random opponent, a hand-built one ("simple rules for scoring and blocking", §6.2), and a **challenger** trained to beat it — his minimax policies held ~35–40% of games against their challenger, his Q-learning (deterministic) policies won **0%**. Reproduced here with the *exact* solver on his 5×4 / 5-action / random-order / goal-reset board (scripts/tournament.py, score = player-0 expected discounted goal difference; the hand-built opponent beats a random one ~76% of games):

LITTMAN'S BOARD, $\Gamma = 0.9$ – PLAYER-0 GOAL DIFFERENCE FROM THE KICKOFF

PLAYER-0 POLICY	VS. RANDOM	VS. HAND-BUILT	VS. ITS CHALLENGER	ROBUSTNESS GAP
minimax (Nash)	+0.67	+0.28	+0.15	+0.52
greedy vs. random	+0.92	+0.68	-0.59	+1.51
greedy self-play	+0.05	0.00	-0.00	+0.05
hand-built	+0.72	0.00	-0.68	+1.39

Minimax holds up; greedy collapses to its challenger



Only minimax is in the top-left. greedy/rand is the strongest policy against weak opponents and among the weakest against a challenger – the rock-paper-scissors trap. minimax presses its advantage (+0.67) and still wins (+0.15) against the worst case, because the Nash policy has no exploitable row.

The mechanism is exactly Littman's "every deterministic offense has a perfect defense": the mixed states are matching-pennies stage games; the greedy *and* the hand-built policy each commit to one row of each, and the challenger plays the column that beats it. The minimax policy randomizes with the weights that make the challenger indifferent. Full table (including the $P(\text{win}) - P(\text{loss})$ view) in docs/tournament.md.

Other checks

- Duality gap $V0_{br}(s_0) + V1_{br}(s_0) < 1e-9$ for every move order — no opponent beats the game value.
- Nash vs. Nash reproduces the value: forced draws in the deterministic and coin-flip games; $+0.162 \pm 0.002$ empirical discounted return over 5 seeds of 3000 games (vs. $+0.164$ computed) in the random game.
- A best response to one assumed opponent is fragile: the Part 2 policy has exploitability 0.43 — worse than random — which is the A10 competition's warning about non-equilibrium submissions.
- A10 networks over 5 seeds (experiments/a10_competition_seeds.csv): the Part 2 imitation net plays optimally at 0.990 ± 0.000 of states and wins Q8 every time; competition Network Second is exact, but Network First sits at a 0.19 ± 0.02 exploitability from this kickoff that does not wash out with re-seeding.

Beyond zero-sum, and open questions

- **General-sum — extension point.** `markov_game.py` solves 2-player general-sum Markov games by enumerating *all* stage equilibria (`support_enum.py`) and selecting one — the notes' "largest sum of values", a no-op for zero-sum but decisive for Battle of the Sexes, where it avoids the mixed equilibrium whose value is below either pure one. Pure-first throughout. The soccer game is zero-sum, so this does not touch the main result.
- **Function approximation — partial.** A network fit to the stage matrices (`nash_dqn.py`, $5 \rightarrow 96 \rightarrow 96 \rightarrow 16$ with biases, 600 epochs) trails the exact solver, over 5 seeds (`experiments/nash_dqn_seeds.csv`):

	EXACT HYBRID NASH-Q	NEURAL NASH-Q (MEAN $\pm$ SD)
max $ V - V_{\text{exact}} $	0	0.55 $\pm$ 0.02
mean $ V - V_{\text{exact}} $	0	0.13 $\pm$ 0.00
action agreement	100%	43% $\pm$ 2%
pure/mixed classification	100%	67% $\pm$ 2%
exploitability	$<1\text{e-}9$	0.44 $\pm$ 0.05
convergence (epochs to plateau, of 600)	exact	469 $\pm$ 31; 3/5 still creeping
runtime	9 sweeps, 0.3 s	600 epochs, ~35 s

Deterministic game only — `train_nash_dqn` rejects the stochastic variants, so the network never has to represent a genuinely mixed stage game. Even so it trails: it fits the ~1000 zero-value states easily (small *mean* error) but misses the γ^k bands and contested regions — worst-state error 0.55, policy about as exploitable as random play, mis-calls *whether* a state needs mixing a third of the time. Keep the exact solver as ground truth.

- **Four research questions** — the single-cell theorem's last step, whether the goal-width dichotomy is a known result, the move-order-vs-randomness distinction, and why freeze-then-iterate thrashes — are stated and answered as far as the evidence allows in `docs/discussion.md`.

Limitations

- The A10 page redacts the ID-specific start position and goal rows; this repo uses the standard Littman geometry (configurable). Across the phase-diagram sweep every one-cell-goal board has 0 mixed states and every wider-goal board has some; the exact 3.95% is the 7×5 / 3-cell-goal case.
 - Several collision sub-cases (carrier vs. stationary opponent, bump = steal, swap possession) are *interpreted* from the two stated A10 rules, not quoted. If course staff intended otherwise, Claim A must be re-measured. See docs/assumptions.md.
 - **Claim A/B** (a pure equilibrium of the deterministic game) is established constructively — the explicit pure memoryless attractor/safety profile. **Claim C** (the theoretical game necessarily has one, over all rules and settings) is *not*, for the general goal mouth; for the single goal cell it is partly proved (docs/proof.md).
 - random and coinflip are different game definitions, isolating the collision rule and the tie-break respectively — not the A10 game.
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soccer_nash — game.py · matrix_games.py · support_enum.py · nash_q.py · markov_game.py ·
numerics.py · dominance.py · attractor.py · geometry.py · symmetry.py · tree.py ·
templates.py · onecell.py · reachability.py · occupancy.py · evaluate.py ·
best_response.py · exploit.py · mlp.py · nash_dqn.py · opponents.py · shaping.py ·
simulate.py · experiment.py · render.py · viz.py
scripts — experiments.py · analyze.py · numerics.py · equilibria.py · geometry_model.py ·
templates.py · phase_diagram.py · onecell_proof.py · positional.py · benchmark.py ·
policy_iteration.py · selfplay.py · nash_dqn.py · render.py · gallery.py · littman.py ·
mixing.py · reward.py · blend.py · occupancy.py · tournament.py · a10_part1.py ·
a10_part2.py · a10_competition.py
docs — design.md · advisor.md · methods.md · discussion.md · numerics.md · assumptions.md
· templates.md · proof.md · geometry.md · mechanism.md · littman.md · tournament.md ·
mixing.md · reward.md · blend.md · occupancy.md · gallery.html
experiments — baseline.csv · undiscounted.csv · gamma_sweep.csv · tolerance_sweep.csv ·
board_sweep.csv · goal_mouth_sweep.csv · one_cell_goal.csv · phase_diagram.csv ·
nash_dqn_seeds.csv
env: 7×5 grid, 2380 states, zero-sum ± 1 , $\gamma = 0.9$